

Thomas Bury

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Last updated: June, 2026
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EDUCATION

- 2015 – 2019 **PhD, Applied Mathematics**, University of Waterloo, Canada
Thesis: Detecting and distinguishing transitions in ecological systems: model and data-driven approaches.
Advisors: Dr. Chris Bauch, Dr. Madhur Anand
- 2014 – 2015 **MMATH, Mathematics**, University of Cambridge, UK
Director of studies: Dr. Julia Gog, OBE
- 2011 – 2014 **BA, Mathematics**, University of Cambridge, UK

PROFESSIONAL APPOINTMENTS

- 2026 – present **Assistant Professor**
Department of Mathematics
University of California, Riverside, USA
- 2020 – 2025 **Postdoctoral Researcher**
Department of Physiology
McGill University, Canada

GRANTS AND FELLOWSHIPS

- 2023 – 2026 (Collaborator) NWO Vidi grant - To tip, or not to tip, that is the question (800,000 euros)
- 2022 – 2025 FRQNT postdoctoral research scholarship (\$135,000)
- 2020 – 2022 CAMBAM postdoctoral fellowship, Centre for Applied Mathematics in Bioscience and Medicine, McGill University (\$17,000)

AWARDS & HONORS

- 2025 Runner-Up, Best Poster Award, Gordon Research Conference on Cardiac Arrhythmia Mechanisms (\$400 USD)
- 2024 Post-Graduate Students' Society Travel Award, McGill University (\$500 CAD)
- 2023 CRBS Travel Award, McGill University (\$1,000 CAD)
- 2021 PNAS Cozzarelli Prize Finalist for Scientific Excellence and Originality
- 2019 Doctoral Thesis Award, University of Waterloo (\$5,000 CAD)
- Combined Travel Grants, Waterloo Institute for Complexity and Innovation (\$2,500 CAD)
- 2017 Research Dissemination Award, GRADTalks, University of Waterloo (\$500 CAD)
- Second Place, Fields Institute Thesis Competition (\$300 CAD)
- Finalist, 3-Minute Thesis Competition, University of Waterloo (\$100 CAD)

PUBLICATIONS

UNDER REVIEW

- Grohgan M, Bury TM, van der Bolt B, Reichart GJ, Hennekam R. An Earth system deep learning classifier for tipping point detection. *EGUsphere*. 2026 Apr 23;2026:1-25.
- Penru NP, Bury TM, Sethi SS, Ewers RM, Picinali L. Prospects for acoustically monitoring ecosystem tipping points. *arXiv preprint arXiv:2509.02201*. 2025 Sep 2.

PEER-REVIEWED JOURNAL ARTICLES

- [1] Tomek J, Holmes M, Bury TM, Tomkova M, Jo H, Nagy N, Bertrand A, Bueno-Orovio A, Colman MA, Rodriguez B, Bers DM. T-world virtual human cardiomyocyte. I: Development, validation, and cell arrhythmogenesis. *Circulation Research*. 2026 May 8;138(10):e328073.
- [2] Diagne K, Bury TM, Pettebone ME, Deyell MW, Laksman Z, Shrier A, Glass L, Bub G, Entcheva E. Phase resetting in human stem cell derived cardiomyocytes explains complex cardiac arrhythmias. *PLOS Computational Biology*. 2026 Feb 4;22(2):e1013935.
- [3] Osakwe A, Wightman N, Deyell MW, Laksman Z, Shrier A, Bub G, Glass L, Bury TM. Dependence of premature ventricular complexes on heart rate—it's not that simple. *Journal of the American Medical Informatics Association*. 2026 Jan;33(1):90-7.
- [4] Ma Z, Zeng C, Zhang YC, Bury TM. Predicting critical transitions with machine learning trained on surrogates of historical data. *Communications Physics*. 2025 Jun 19;8(1):258.
- [5] Sadria M, Bury TM. FateNet: an integration of dynamical systems and deep learning for cell fate prediction. *Bioinformatics*. 2024 Sep;40(9):btae525.
- [6] Bury TM, Diagne K, Olshan D, Glass L, Shrier A, Lerman BB, Bub G. The inverse problem for cardiac arrhythmias. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2023 Dec 1;33(12).

- [7] Bury TM, Dylewsky D, Bauch CT, Anand M, Glass L, Shriner A, Bub G. Predicting discrete-time bifurcations with deep learning. *Nature Communications*. 2023 Oct 10;14(1):6331. (**Editor’s Highlight**)
- [8] Dylewsky D, Lenton TM, Scheffer M, Bury TM, Fletcher CG, Anand M, Bauch CT. Universal early warning signals of phase transitions in climate systems. *Journal of the Royal Society Interface*. 2023 Apr 1;20(201).
- [9] Bury TM. ewstools: A Python package for early warning signals of bifurcations in time series data. *Journal of Open Source Software*. 2023 Feb 11;8(82):5038.
- [10] Diagne K, Bury TM, Deyell MW, Laksman Z, Shriner A, Bub G, Glass L. Rhythms from two competing periodic sources embedded in an excitable medium. *Physical Review Letters*. 2023 Jan 13;130(2):028401. (**Editor’s Suggestion**)
- [11] Bury TM, Sujith RI, Pavithran I, Scheffer M, Lenton TM, Anand M, Bauch CT. Deep learning for early warning signals of tipping points. *Proceedings of the National Academy of Sciences*. 2021 Sep 28;118(39):e2106140118. (**Cozzarelli finalist**)
- [12] Menard J, Bury TM, Bauch CT, Anand M. When conflicts get heated, so does the planet: coupled social-climate dynamics under inequality. *Proceedings of the Royal Society B: Biological Sciences*. 2021 Sep 1;288(1959).
- [13] Bury TM, Lerma C, Bub G, Laksman Z, Deyell MW, Glass L. Long ECGs reveal rich and robust dynamical regimes in patients with frequent ectopy. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2020 Nov 1;30(11).
- [14] Bury TM, Bauch CT, Anand M. Detecting and distinguishing tipping points using spectral early warning signals. *Journal of the Royal Society Interface*. 2020 Sep 1;17(170).
- [15] Bury TM, Bauch CT, Anand M. Charting pathways to climate change mitigation in a coupled socio-climate model. *PLOS Computational Biology*. 2019 Jun 6;15(6):e1007000.
- [16] Pananos AD, Bury TM, Wang C, Schonfeld J, Mohanty SP, Nyhan B, Salathé M, Bauch CT. Critical dynamics in population vaccinating behavior. *Proceedings of the National Academy of Sciences*. 2017 Dec 26;114(52):13762-7.

LETTERS AND COMMENTARIES

- [1] Dablander F, Bury TM. Deep learning for tipping points: preprocessing matters. *Proceedings of the National Academy of Sciences*. 2022;119(37):e2207720119.

SELECTED MEDIA COVERAGE

- 2023 “Signatures of Competing Cardiac Pacemakers”, APS Physics
“The subtle math of a heartbeat gone wrong”, Physics Today
- 2021 “Artificial intelligence could reveal climate change tipping points”, The Independent
- 2019 “Changes in individual human behaviour can help mitigate climate change”, The Globe and Mail
- 2017 “Researchers predict vaccine scares using Google and Twitter trends”, CBC News

OPEN-SOURCE SOFTWARE

2019 – present **ewstools**

A Python package for computing early warning signals for bifurcations in time series data

- Role: Creator, core developer
- GitHub:  [ThomasMBury/ewstools](https://github.com/ThomasMBury/ewstools), 90+ stars

2025 – present **T-World GUI**

Interactive graphical user interface for the T-World human ventricular myocyte model.

- Role: Lead developer
- GitHub:  [ThomasMBury/t-world-simulator-multipage](https://github.com/ThomasMBury/t-world-simulator-multipage)

RESEARCH VISITS AND SCIENTIFIC PROGRAMS

2026

Invited Scientist, Kavli Institute for Theoretical Physics (KITP)

Program: Multi-Scale Physics of Normal and Diseased Heart: from Ion Channels to Whole Organ

University of California, Santa Barbara

5–24 April (3 weeks)

INVITED SEMINARS

2026

Kavli Institute for Theoretical Physics (KITP), University of California, Santa Barbara, Santa Barbara, CA, USA (two invited seminars, 11 Apr and 22 Apr)

ICDMB Seminar Series, University of California, Riverside, Riverside, CA, USA, 24 Feb

2025

McMaster University, Dept. of Mathematics, Hamilton, ON, Canada, 10 Apr

University of California, Riverside, Dept. of Mathematics, Riverside, CA, USA, 14 Jan

University of Houston, Dept. of Mathematics, Houston, TX, USA, 13 Jan

2024

Rowan University, Dept. of Mathematics, Glassboro, NJ, USA, 17 Dec

University of South Carolina, Dept. of Mathematics, Columbia, SC, USA, 2 Dec

Montreal Institute for Learning Algorithms (MILA), Montréal, QC, Canada, 31 May

Waterloo Institute for Complexity and Innovation (WICI), Waterloo, ON, Canada, 2 May

Centre de Recherche en Biologie Structurale Seminar Series, McGill University, 22 March

2021

Department of Physiology Seminar Series, McGill University, 22 January

2020

University of Ottawa, Dept. of Applied Mathematics, 26 November

Seminar Series in Quantitative Life Sciences and Medicine, McGill University, 22 September

Centre de Recherches Mathématiques (CRM), Université de Montréal, 11 May

CONFERENCE PRESENTATIONS

- 2025 Gordon Research Conference on Cardiac Arrhythmia Mechanisms, Lucca, Italy, 23–28 February (poster)
- 2024 SIAM Conference on Mathematics of Data Science, Atlanta, GA, USA, 21–25 October (**Mini-Symposium Organizer**)
- SIAM Annual Meeting, Spokane, WA, USA, 8–12 July (**Invited Mini-Symposium Talk**)
- 2023 Workshop: Challenges of predicting critical transitions in natural systems, University of Exeter, UK, 7–8 December (**Invited Plenary**)
- SIAM Conference on Applications of Dynamical Systems, Portland, OR, USA, 14–18 May (**Invited Mini-Symposium Talk**)
- 2021 Dynamics Days Europe, Virtual, 23–27 August
- Society for Mathematical Biology Annual Meeting, Virtual, 13–17 June
- 2020 Workshop on Critical Transitions in Complex Systems, Shanghai Institute for Biological Sciences, Virtual, 29–31 July (**Invited Speaker**)
- 2019 Society for Mathematical Biology Annual Meeting, Montréal, QC, Canada, 22–26 July (poster)
- Canadian Society of Applied and Industrial Mathematics Annual Meeting, Whistler, BC, Canada, 9–13 June
- 2018 Ecological Society of America Annual Meeting, New Orleans, LA, USA, 5–10 August
- Dynamics Days US, Denver, CO, USA, 4–6 January
- 2017 Applied Mathematics, Modeling and Computational Science, International Conference, Waterloo, ON, Canada, 20–25 August
- Mathematical Models in Ecology and Evolution, Conference, London, UK, 10–12 July
- WICI Interdisciplinary Conference on Resilience in Complex Natural and Human Systems, Waterloo, Canada, 16–17 May

TEACHING

GRADUATE

- 2021 – 2022 **Instructor**, *McGill University*
 QLSC 600D1: Foundations of Quantitative Life Sciences (Module: Resetting and Entraining Biological Rhythms; 15–20 students; Fall 2021, Fall 2022)
- 2017 – 2018 **Teaching Assistant**, *University of Waterloo*
 AMATH 777: Stochastic Processes in the Physical Sciences (Winter 2017, Winter 2018)
 MATH 650: Mathematical Modeling with Differential Equations (Fall 2017)

UNDERGRADUATE

- 2026 **Instructor**, *University of California, Riverside*
 MATH 007B: Calculus for Life Sciences II (120 students), Winter 2026
 MATH 046: Introduction to Ordinary Differential Equations (92 students), Winter 2026
- 2024 **Guest Lecturer**, *McGill University*
 BIOL 309: Mathematical Models in Biology (80 students, Fall 2024)
- 2018 **Instructor**, *University of Waterloo*
 MATH 127: Calculus I for the Sciences (110 students, Fall 2018)
- 2015 – 2018 **Teaching Assistant**, *University of Waterloo*
 AMATH 353 (PDEs), AMATH 350 (Differential Equations for Business and Economics),
 AMATH 250 (Introduction to Differential Equations), MATH 116 (Calculus I)

SUMMER SCHOOLS AND WORKSHOPS

- 2021 **Technical lead**, Summer School in Nonlinear Dynamics for the Life Sciences, McGill University (virtual), 31 May - 11 June
- 2020 **Workshop creator and facilitator**, Interactive Data Visualisation in Python, McGill University (virtual), 27 July
- 2018 **Workshop co-creator and co-facilitator**, A Hands-on Introduction to Mathematical Modelling, Waterloo Institute for Complexity and Innovation, 26 April

TEACHING CREDENTIALS

- 2017 – 2019 Certificate of University Teaching, *University of Waterloo*
- 2015 – 2016 Fundamentals of University Teaching, *University of Waterloo*

MENTORSHIP AND ADVISING

DOCTORATE

- 2019 – 2025 Khady Diagne (Research Mentor)
 McGill University

MASTERS

- 2019 – 2020 Jyler Menard (Research Mentor)
 University of Waterloo

UNDERGRADUATE

- 2026 – present Chao Wei Hsu (Supervisor)
University of California, Riverside
- 2020 – 2021 Alix Vanpoperinghe (Supervisor)
McGill University
- 2020 – 2021 Glisant Plasa (Co-supervisor)
McGill University

ACADEMIC SERVICE

DEPARTMENTAL

- 2026 – present Member, Computational Lab Committee, Department of Mathematics, University of California, Riverside
- 2026 Co-host, PDE & Applied Mathematics Seminar, Department of Mathematics, University of California, Riverside (Winter 2026)

CONFERENCE ORGANIZATION

- 2024 Organizer, Mini-Symposium “Early Warning Signals in Medicine”, SIAM Conference on Mathematics of Data Science

PROFESSIONAL

- 2025 Member, Mackenzie King Scholarships Selection Committee, *McGill University*
- 2024 Member, EDIA Champions Merit Review Committee, *Digital Research Alliance of Canada*
- 2021 – 2025 Evaluation Committee Member, Tri-Agency Canada Graduate Scholarship–Master’s Competition, *McGill University*
- 2017 – 2018 Graduate Student Representative, Senate Graduate and Research Council, *University of Waterloo*

OUTREACH

- 2026 Science Fair Judge, Riverside Unified School District Science & Engineering Fair, Riverside, CA
- 2024 Career Panelist, McGill Quantitative Life Sciences Research Day
Invited Speaker, IFMSA-Quebec Global Health Symposium, Montréal, QC, Canada
- 2023 Lecture to CEGEP Students, Youreka Canada, “Data Science: Practice and Principles”, Montréal, QC, Canada
- 2022 Lecture to Secondary School Students, Kelly College, “Mathematics Beyond School: University, Careers and Life”, Devon, UK
- 2021 – 2024 Technical Author, Towards Data Science

2021 Media Interviews, *The Charlatan* and *The McGill Tribune*
2016 – 2018 Workshop Facilitator, *Let's Talk Science*, Waterloo, ON, Canada
2017 TEDx Speaker, University of Toronto, Toronto, ON, Canada

AD HOC PEER REVIEW

Nature, Nature Machine Intelligence, Nature Climate Change, Nature Communications, PNAS, Physical Review Letters, Physical Review X, Ecology Letters, Royal Society journals, Chaos, Journal of Theoretical Biology, Physica D, iScience, PLOS Computational Biology, npj Systems Biology and Applications.

PROFESSIONAL MEMBERSHIPS

Society for Industrial and Applied Mathematics (SIAM)
Canadian Applied and Industrial Mathematics Society (CAIMS)

LANGUAGES

English Native
French Proficient : TEFaQ Level C1 obtained in 2020